ISS 2103 • EXAM REVISION PACK 03

Physical Layer: Media, Signals, Multiplexing & Switching

Course outline weeks 1–2: transmission media, switching methods, ISDN, terminal handling

Priority: **VERY HIGH**

Physical-layer questions appear in **every** paper, usually two or three parts per paper (2–9 marks each). Switching has come up in 4 of the 5 papers, and media in 4 of 5. These are cheap, predictable marks: learn the tables and the diagrams.

How often this topic is examined

Question asked	Paper(s)	Marks
Circuit vs packet (vs message) switching; examples; latency & jitter	2023 Q5b.i–iii, ISA2024 Q2d.i–ii, SE2024 Q2c.i–ii, 2025 Q3a	2–9
Types of transmission media (list / explain / enterprise, with diagram)	2023 Q1e, 2025 Q2b.i, SE2024 Q4c	4
Bounded vs unbounded media	SE2024 Q1b.i	2
Fibre optic: how it transmits (diagram) + why it beats copper	2023 Q3a, Q3b	5–6
Multiplexing: concept & importance; MUX components; FDM vs TDM	2025 Q2b.ii, SE2024 Q1b, Q2a	4–5
Transmission impairments (attenuation, distortion, noise)	SE2024 Q1d	3
Analogue vs digital signals; synchronous transmission	SE2024 Q3c.ii, Q1a.iii	1–2
Modulation vs encoding, with examples	SE2024 Q5b	6
ADSL: why asymmetric; voice + data on one pair	ISA2024 Q5b.i–ii	2–5
GSM vs CDMA	SE2024 Q1b.iii	3
Nyquist / Shannon, ISDN, terminal handling, WDM	not asked yet (in outline)	–

EXAM TIP

The ISA papers (2024, 2025) asked switching *every* year and ADSL once. Have the circuit/message/packet table and the three-network diagram ready to draw in under five minutes. “Using a diagram” in the question means the diagram itself carries about half the marks.

1 Transmission media

The **transmission medium** is the physical path between transmitter and receiver (the “layer 0” under the physical layer).

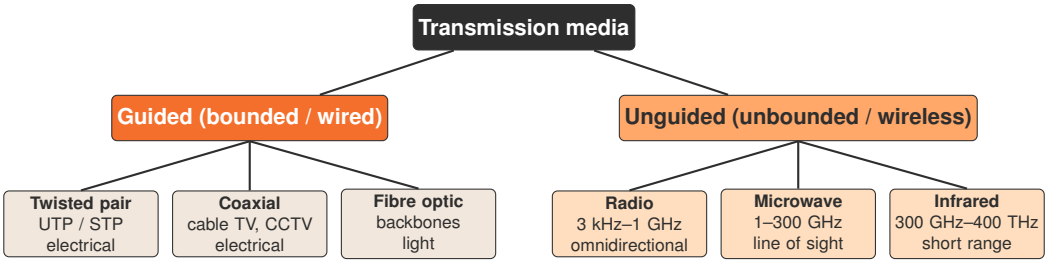


Figure: classification of transmission media (lecturer’s terms: guided = bounded, unguided = unbounded).

1.1 Copper media

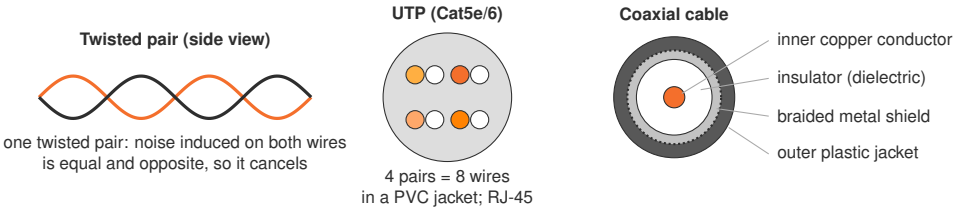


Figure: twisted pair (twisting cancels interference) and coax (the shield blocks outside interference).

- **Twisted pair**: pairs of insulated copper wires twisted together. The two wires carry + and − signals (balanced line); noise hits both equally and cancels at the receiver. **UTP** (unshielded, cheap, Ethernet LANs, max ≈100 m) and **STP** (foil shield, better against EMI and crosstalk).
- **Coaxial**: two conductors sharing one axis: inner conductor, insulator, braided outer conductor (shield), jacket. Better shielding and higher bandwidth than UTP. Cable TV, CCTV, old 10Base2/10Base5 Ethernet.
- **Fibre optic**: glass core carries **light**; see Section 2.
- **Radio** (omnidirectional, penetrates walls: AM/FM, Wi-Fi), **microwave** (unidirectional, line of sight, dish antennas, repeaters/towers, satellite), **infrared** (cannot pass walls, short range: remotes, IrDA).

	Twisted pair	Coaxial	Fibre optic
Signal	Electrical	Electrical	Light
Bandwidth	Low–medium (100 MHz–1 GHz)	Medium	Very high (THz)
Distance	≈100 m	≈500 m	2–100+ km
Noise / EMI	Susceptible (STP better)	Less (shielded)	Immune
Cost / install	Cheapest, easy	Moderate	Expensive, needs skilled splicing
Use	LAN to the desk, phone line	Cable TV, CCTV	Backbones, WAN, undersea

PAST PAPER **2023 Q1e / 2025 Q2b.i** **[4 marks]**

List any four (4) types of transmission media that you are familiar with. / *The Physical Layer is responsible for the transmission of bits over a physical medium.* Explain two types of transmission media.

Model answer

List (2023, 1 mark each): twisted pair (UTP/STP), coaxial cable, fibre optic cable, radio waves (Wi-Fi), microwave/satellite, infrared.
Explain two (2025, 2 marks each):

1. **Guided – twisted pair**: insulated copper wires twisted in pairs so induced noise cancels; carries electrical signals; cheap and easy to install; used for Ethernet LANs (RJ-45, Cat6), limited to about 100 m and affected by EMI.
2. **Unguided – radio / microwave**: data carried as electromagnetic waves through air with no cable; radio is omnidirectional (Wi-Fi, mobile), microwave needs line of sight between dishes; gives mobility and easy installation over difficult terrain but is less secure and affected by weather and interference.

PAST PAPER **SE2024 Q1b.i** **[2 marks]**

Outline the difference between Bounded and unbounded transmission media.

"Explain **two types**" is safest as one *guided* and one *unguided* example: it shows you know both families.

Model answer

Bounded (guided)	Unbounded (unguided)
Signal is confined to a physical conductor (copper or glass) along a fixed path	Signal travels through air/vacuum as electromagnetic waves, no conductor
E.g. twisted pair, coax, fibre	E.g. radio, microwave, infrared, satellite
More secure, less interference, capacity added by adding cables	Mobility, cheap to install over distance; less secure, weather/interference

PAST PAPER **SE2024 Q4c** **[4 marks]**

Using a diagram, explain the features of the types of networking media used in enterprise networks.

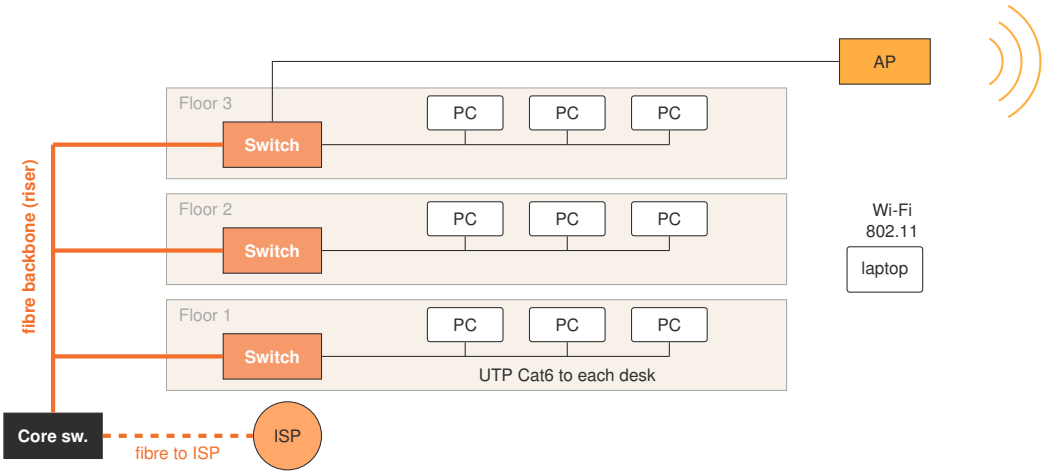


Figure: structured cabling. Fibre between floors and to the ISP, UTP to desks, Wi-Fi for mobile users.

Model answer

1. **UTP Cat5e/Cat6 (horizontal cabling)**: from the floor switch to each desk; cheap, easy to terminate (RJ-45), up to 1–10 Gbps over $\leq$100 m.
2. **Fibre optic (backbone / vertical riser)**: links floor switches to the core switch and the building to the ISP; very high bandwidth, long distance, immune to EMI, secure.
3. **Wireless (Wi-Fi access points)**: unguided radio for laptops and phones; mobility and no cabling, but shared bandwidth and needs WPA2/3 security.
4. (Coax, if mentioned: CCTV and some ISP cable-modem links.)

2 Fibre optic cable

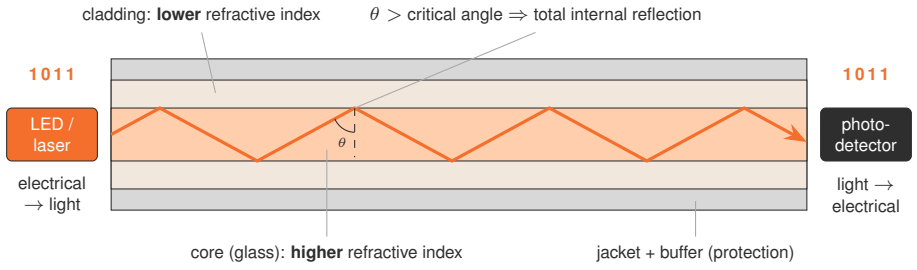


Figure: light pulses bounce along the core by total internal reflection (light on = 1, off = 0).

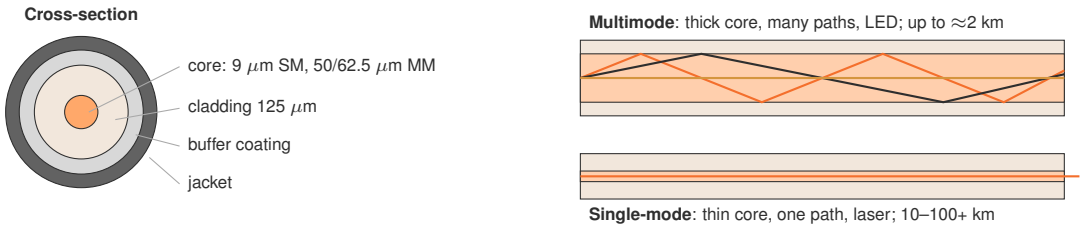


Figure: fibre cross-section, and multimode vs single-mode light paths.

PAST PAPER **2023 Q3a** **[5 marks]**

Using a diagram, explain how data is transmitted along a fiber optic cable.

Model answer

(Draw the light-path figure above, then:)

1. At the transmitter, the electrical bits are converted to **light pulses** by an LED or laser diode (light on = 1, off = 0).
2. The light enters the glass **core**, which has a **higher refractive index** than the surrounding **cladding**.
3. Rays hit the core/cladding boundary at more than the **critical angle**, so they undergo **total internal reflection** and bounce along the core with very little loss, even round bends.
4. The buffer and jacket protect the fibre from damage and moisture.
5. At the far end a **photodetector** (photodiode) converts the light pulses back to electrical signals.

PAST PAPER **2023 Q3b** **[6 marks]**

Identify three (3) physical characteristics of fiber optic cables that make them more suitable for high-speed digital data transmission over copper cables.

Key phrase for the mark: **total internal reflection**, because core index > cladding index and angle > critical angle.

Model answer

1. **Very high bandwidth**: light has a frequency of hundreds of THz, so fibre carries Gbps–Tbps, and WDM puts many wavelengths on one fibre.
2. **Low attenuation**: signals travel tens of km without repeaters (UTP needs one every 100 m).
3. **Immune to EMI and crosstalk**: it carries light, not electricity, so motors, lightning and nearby cables add no noise, giving very low error rates.

Also: thin and light (many fibres per cable), hard to tap without detection (security), no sparks or ground loops (electrical isolation).

TRAP – where students lose marks

The core has the **higher** refractive index, the cladding the **lower**. Reversing them is the most common error. Also: it is *total internal reflection*, not refraction.

3 Signals, impairments and data rate

3.1 Analogue vs digital; synchronous transmission

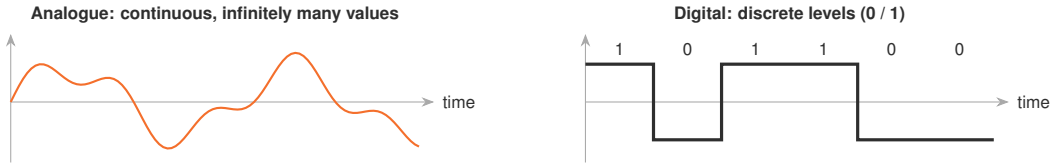


Figure: analogue vs digital signal.

PAST PAPER **SE2024 Q3c.ii** **[2 marks]**

Distinguish between the following terms: Analogue verses digital signals.

Model answer

Analogue signal	Digital signal
Continuous wave; infinitely many values in a range	Discrete; a limited number of levels (e.g. 0 and 1)
Described by amplitude, frequency, phase	Described by bit rate and bit interval
Noise is added and cannot be removed (amplifiers boost it too)	Can be regenerated exactly by repeaters; more noise-immune
E.g. human voice, analogue telephone, AM/FM radio	E.g. computer data, Ethernet

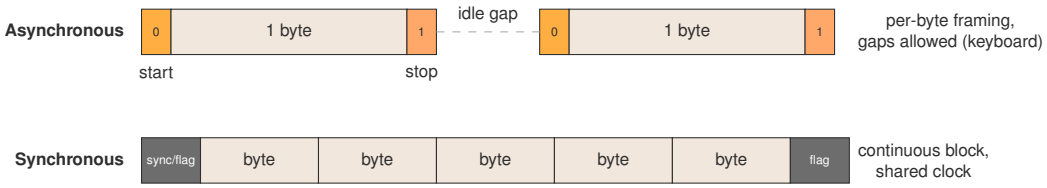


Figure: asynchronous (start/stop bits per byte) vs synchronous (one continuous clocked block).

PAST PAPER **SE2024 Q1a.iii** **[1 marks]**

Explain the following term: Synchronous transmission.

Model answer

Data is sent as a **continuous stream of bits in blocks/frames**, with sender and receiver kept in step by a **common clock** (timing signal); there are no start/stop bits or gaps per byte, so it is faster and more efficient than asynchronous. E.g. Ethernet, HDLC frames.

3.2 Transmission impairments

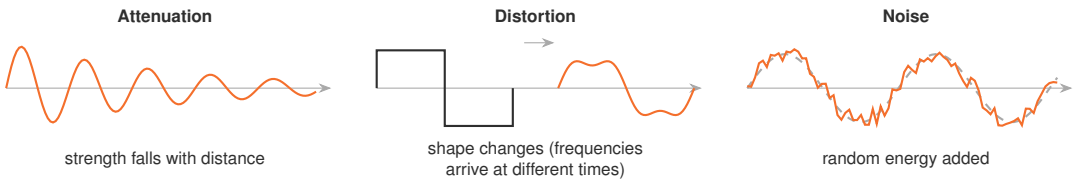


Figure: the three transmission impairments (sent signal grey/black, received signal orange).

PAST PAPER **SE2024 Q1d** **[3 marks]**

Signals picks unwanted random energy, discuss such three transmission impairments.

Model answer

1. **Attenuation**: loss of signal energy as it travels, overcoming the resistance of the medium (energy lost as heat). Measured in dB; fixed with amplifiers (analogue) or repeaters (digital).
2. **Distortion**: the signal changes shape because its frequency components travel at different speeds and arrive at different times (delay distortion; dispersion in fibre).
3. **Noise**: unwanted random energy added to the signal: **thermal** (random electron motion), **induced** (motors, appliances), **crosstalk** (one wire picks up another) and **impulse** (spikes

from lightning, power lines). Quality is measured by the signal-to-noise ratio (SNR).

3.3 Bandwidth and maximum data rate (outline topic)

The wording "unwanted random energy" describes *noise*. If unsure, give attenuation/distortion/noise **and** name the four noise types: it covers both readings.

- **Bandwidth** (Hz) = the range of frequencies a channel passes, $B = f_{\text{high}} - f_{\text{low}}$. **Bit rate** = bits per second. **Baud rate** = signal elements per second; bit rate = baud $\times \log_2 L$.
- **Nyquist** (noiseless channel): BitRate = $2B \log_2 L$, L = number of signal levels.
- **Shannon** (noisy channel): $C = B \log_2(1 + \text{SNR})$, SNR as a ratio, not dB. $\text{SNR}_{\text{dB}} = 10 \log_{10} \text{SNR}$.

Lecture-notes examples

Nyquist: $B = 5000$ Hz, $L = 2$

$\text{BitRate} = 2 \times 5000 \times \log_2 2 = 10\,000$ bps ← $\log_2 2 = 1$

Shannon: $B = 3000$ Hz, SNR = 300

$C = 3000 \times \log_2(301)$ ← $\log_2 x = \ln x / \ln 2$

$= 3000 \times 8.23 \approx 24\,700$ bps ← $\ln 301 / \ln 2 = 5.707/0.693$

dB first: $\text{SNR}_{\text{dB}} = 30 \Rightarrow \text{SNR} = 10^{30/10} = 1000$ ← *always convert dB to a ratio*

TRAP – where students lose marks

Putting SNR in dB straight into Shannon. $C = 3000 \log_2(1 + 30)$ for a 30 dB line is **wrong**; use SNR = 1000. Nyquist gives the limit for a *chosen* number of levels; Shannon gives the absolute ceiling no matter how many levels you use.

Calculator trick: $\log_2 x = \log_{10} x \div 0.301$. So $\log_2 301 = 2.479/0.301 = 8.23$.

4 Modulation and line coding

Modulation (digital-to-analogue) varies one property of a sine **carrier** to represent bits: **ASK** (amplitude), **FSK** (frequency), **PSK** (phase), **QAM** (amplitude + phase). Analogue-to-analogue: AM, FM, PM. **Line coding** / **encoding** (digital-to-digital) maps bits to voltage levels on the wire: NRZ, RZ, Manchester, Differential Manchester, AMI.

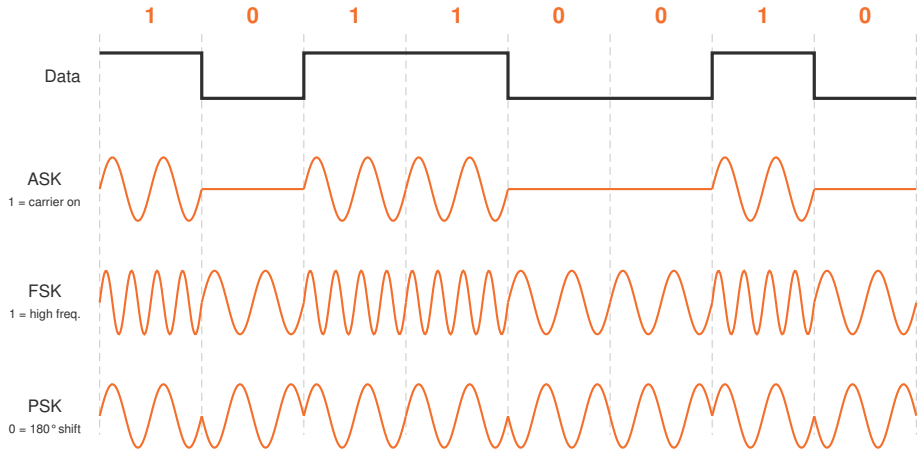


Figure: bit pattern 10110010 sent with ASK (amplitude), FSK (frequency) and BPSK (phase). Only one property changes each time.

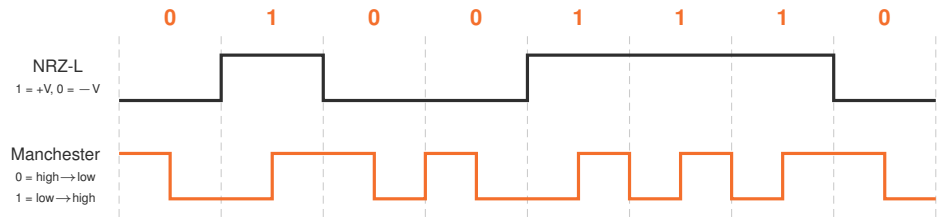


Figure: line coding of 01001110. Manchester (IEEE 802.3) has a transition in the middle of every bit, so the clock is carried in the signal.

PAST PAPER **SE2024 Q5b** **[6 marks]**

Distinguish between modulation techniques verses encoding. Give two examples of each.

Model answer

Modulation	Encoding (line coding)
Converts digital (or analogue) data into an **analogue** signal by varying a carrier's amplitude, frequency or phase	Converts digital data into a **digital** signal: a pattern of voltage levels/transitions
For bandpass channels: telephone lines, radio, long distance	For baseband: short copper links, LAN cables
Device: modem (modulator–demodulator)	Device: NIC / line encoder
Examples: ASK, FSK, PSK, QAM (analogue: AM, FM)	**Examples**: NRZ-L/NRZ-I, Manchester (Ethernet), Differential Manchester, AMI

6 marks = 2 for the definitions, 2 for the contrast, 2 for examples. Sketching ASK and Manchester for 3–4 bits makes it unarguable.

5 Multiplexing

Multiplexing

Combining several signals from several sources into **one composite signal** so they share a **single link**. A **multiplexer (MUX)** is many-to-one; a **demultiplexer (DEMUX)** at the far end is one-to-many and delivers each signal to its own output.

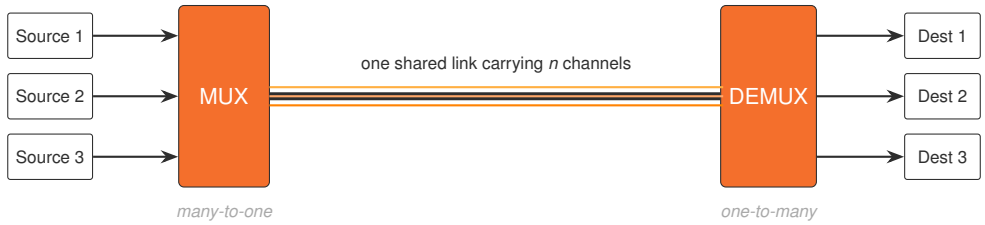


Figure: *n* input lines share one link through a MUX/DEMUX pair.

PAST PAPER **2025 Q2b.ii** **[4 marks]**

Describe the concept of multiplexing and its importance in data communication.

Model answer

Concept: a MUX combines *n* signals into one composite signal on a single link; the DEMUX separates them at the receiver. The link is divided into channels by frequency (FDM), time (TDM) or wavelength of light (WDM).
Importance:

1. **Efficient use of bandwidth**: one high-capacity link carries many low-rate signals instead of

being under-used.
2. **Lower cost**: one cable/fibre instead of many separate lines; less infrastructure to install and maintain.
3. Allows many users to share expensive long-haul links: telephone trunks (E1), cable TV, fibre backbones, ADSL.

PAST PAPER **SE2024 Q1b** **[4 marks]**

Describe the two components that make up a multiplexor.

Model answer

1. **Multiplexer (MUX)** at the sending end: *many-to-one*. It takes *n* input lines and combines them into a single composite stream (assigning each a frequency band or time slot) for the shared link.
2. **Demultiplexer (DEMUX)** at the receiving end: *one-to-many*. It separates the composite stream back into the *n* original signals and sends each to its correct output line.

They are joined by one **link** (the physical path) divided into *n* **channels** (the logical paths).

The question is ambiguous. "MUX and DEMUX" is the expected pair; mention "link and channels" in one line too, to cover the other reading.

5.1 FDM, TDM and WDM

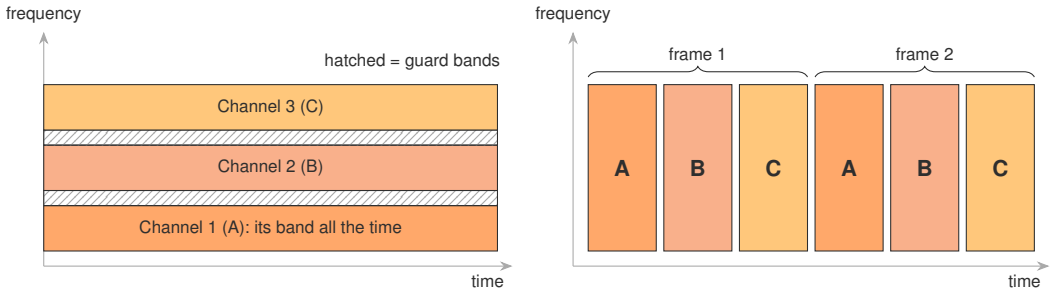


Figure: FDM gives each source a band all the time; TDM gives each source the whole bandwidth for its time slot in turn.

PAST PAPER **SE2024 Q2a** **[5 marks]**

Differentiate between Frequency Division Multiplexing (FDM) and Time Division multiplexing (TDM) techniques.

Model answer

	FDM	TDM
Divides	Link **bandwidth** into frequency bands	Link **time** into slots (frames)
Sharing	Each signal has its own band *all the time*; all sent simultaneously	Each signal uses the *whole* bandwidth for its slot, in rotation
Signal type	Mainly analogue	Mainly digital
Separation	Guard bands between channels	Synchronisation/framing bits, guard time
Problem	Crosstalk between adjacent bands	Empty slots wasted (synchronous TDM) → statistical TDM
Examples	Radio/TV broadcasting, cable TV, ADSL	Digital telephony T1/E1, GSM time slots

- **Synchronous TDM**: each input owns a fixed slot in every frame, even when idle. **Statistical (asynchronous) TDM**: slots are given only to inputs that have data, each slot carries an address.

Better use of the link.

- **WDM** (Wavelength Division Multiplexing): FDM for light. Each signal is a different wavelength ("colour") on the same fibre; a prism/grating combines and splits them. DWDM puts 80+ channels on one fibre.
- **CDM/CDMA**: all users share the same band at the same time, separated by unique codes (spread spectrum).

TDM and FDM sizing

FDM: 5 voice channels of 4 kHz with 1 kHz guard bands. **Sync TDM**: 4 inputs at 64 kbps, 1 byte per slot.

$$B_{\text{FDM}} = 5(4) + 4(1) = 24 \text{ kHz} \quad \leftarrow n \text{ channels need } n-1 \text{ guards}$$

$$\text{TDM frame} = 4 \times 8 = 32 \text{ bits}$$

$$\text{frames/s} = 64\,000/8 = 8000 \quad \leftarrow \text{one byte per input per frame}$$

$$\text{link rate} = 32 \times 8000 = 256 \text{ kbps} \quad \leftarrow = 4 \times 64 \text{ kbps (+ framing bits)}$$

5.2 Multiple access in mobile networks

PAST PAPER **SE2024 Q1b.iii** **[3 marks]**

Compare and contrast Global system for mobile communication (GSM) and Code division multiple access (CDMA) wireless technology.

Model answer

GSM	CDMA
FDMA + TDMA: 200 kHz carriers, each split into 8 time slots (8 users)	All users share the same wide band (1.25 MHz) at the same time, separated by unique codes (spread spectrum)
SIM card: easy to change handset/network	Early systems tied the number to the handset (no SIM)
Hard handoff (break before make)	Soft handoff (make before break): fewer dropped calls
Most of the world (Zimbabwe: Econet, NetOne)	Mainly USA (Verizon, Sprint), parts of Asia

Similarities: both are 2G/3G digital cellular multiple-access systems using cells and base stations; both carry voice and data and were replaced by 4G LTE (OFDMA).

6 Switching

Switching

Directing data from source to destination through intermediate **switching nodes** (switches, routers, exchanges) when there is no direct link between every pair of devices. It avoids the cost of a full mesh.

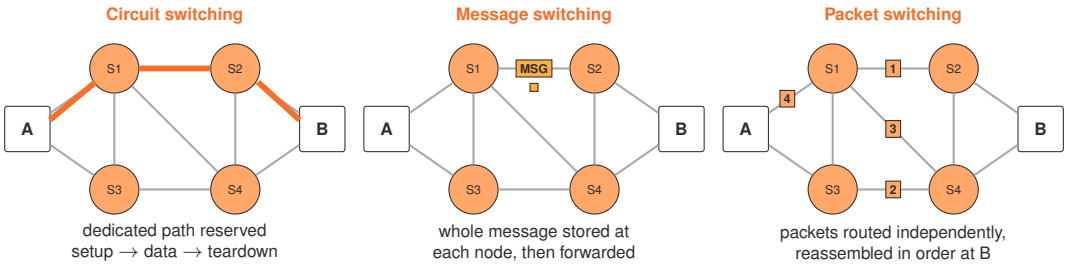


Figure: circuit (reserved path), message (whole message hops node to node) and packet switching (packets may take different routes).

PAST PAPER **2025 Q3a** **[9 marks]**

Explain the concept of switching in networks. Differentiate between message switching, packet switching, and circuit switching.

Model answer

Concept (2–3 marks): switching connects many devices through intermediate switching nodes instead of a direct link between every pair. Each node decides which outgoing link to send the data on. Three methods: circuit, message and packet switching.

	Circuit switching	Message switching	Packet switching
Path	Dedicated end-to-end path set up before transfer, held for the whole call	No dedicated path; whole message stored at each node, then forwarded	No dedicated path; data split into packets, each forwarded independently (datagram) or along a virtual circuit
Phases	Setup → data → teardown	Store-and-forward of whole message	Store-and-forward of small packets
Resource use	Bandwidth reserved even when idle (wasteful)	Better link use, but nodes need large storage	Very efficient: links shared by many users
Delay	Setup delay, then constant low delay, no jitter	High delay (whole message waits at every node)	Variable queuing delay, jitter; packets may arrive out of order
Failure	Link failure drops the call	Message rerouted	Packets rerouted around failures
Example	PSTN telephone, ISDN	Telegraph, early e-mail (telex)	Internet (IP), X.25, Frame Relay

PAST PAPER **2023 Q5b.i / ISA2024 Q2d.i** **[4 marks]**

Switching, as applied to networking and IT, is the practice of directing a signal or data element toward a particular hardware destination. Explain the main difference between a circuit-switched network and a packet switched network.

9 marks: about 2 for the concept, about 2 per method, and the rest for the contrast. The three-network figure earns its marks fast.

Model answer

- **Circuit-switched**: a **dedicated end-to-end path** with fixed bandwidth is set up *before* communication (call setup) and reserved for the whole session, even when nobody is talking; then it is torn down. Data follows the same path, in order, with constant delay.
- **Packet-switched**: **no path is reserved**. Data is split into packets with headers (destination address, sequence number) that share links with other traffic and are forwarded hop by hop (store-and-forward), possibly along different routes, then reassembled at the destination.

Main difference: reserved, dedicated resources (circuit) vs shared, on-demand resources (packet).

PAST PAPER **2023 Q5b.ii / ISA2024 Q2d.ii** **[4 marks]**

Give two (2) examples of packet-switched protocols and two (2) examples of circuit switched protocols.

Model answer

Packet-switched: IP (IPv4/IPv6), X.25, Frame Relay, ATM (cell switching), MPLS, Ethernet.
Circuit-switched: ISDN, PSTN/POTS (SS7 signalling), GSM 2G voice (circuit-switched calls), T1/E1 TDM lines, SONET/SDH.

PAST PAPER **2023 Q5b.iii** **[2 marks]**

Indicate which one would have higher latency and jitter and explain why.

Model answer

Packet switching. Each packet waits in router queues of different lengths, may take a different route, and is stored-and-forwarded at every hop, so delay changes from packet to packet (jitter). A circuit, once set up, has a fixed dedicated path, so its delay is constant.

PAST PAPER **SE2024 Q2c** **[2+2 marks]**

Describe the following switching techniques as used in data communication and networking: (i) Circuit switching (ii) Packet Switching.

Model answer

(i) **Circuit switching**: a dedicated physical path is established between the two stations for the whole duration of the communication (setup, transfer, teardown), e.g. a telephone call.
(ii) **Packet switching**: the message is broken into small packets, each with a header, that travel independently through shared links (store-and-forward) and are reassembled at the destination, e.g. the Internet.

6.1 Datagram vs virtual circuit (two kinds of packet switching)

	Datagram (connectionless)	Virtual circuit (connection-oriented)
Setup	None; send immediately	Route set up first by a call-request packet
Routing	Each packet routed independently; may take different routes	All packets follow the same pre-set route
Header	Full destination address in every packet	Short virtual-circuit identifier (VCI)
Order	May arrive out of order or be lost	Arrive in order
Failure	Avoids failed nodes and congestion	A node failure breaks the circuit
Best for	Short transactions (bank queries)	Long transfers (large files)
Example	IP, UDP traffic	X.25, Frame Relay, ATM

REMEMBER IT

Switching = how a path is found. **Multiplexing** = how one link is shared. Circuit = phone, Message = telegraph, Packet = Internet. FDM = frequency bands (radio), TDM = time slots (E1), WDM = colours of light (fibre).

7 ADSL, ISDN and terminal handling

PAST PAPER **ISA2024 Q5b.i** **[2 marks]**

A common method for providing broadband Internet access over existing telephone lines is Asymmetric Digital Subscriber Line (ADSL). Explain why is it called asymmetric?

Model answer

Because the **downstream** (download) data rate is much higher than the **upstream** (upload) rate, e.g. up to 8 Mbps down vs about 1 Mbps up (ADSL2+: 24 Mbps down). More of the frequency spectrum is given to downstream, because home users download far more than they upload.

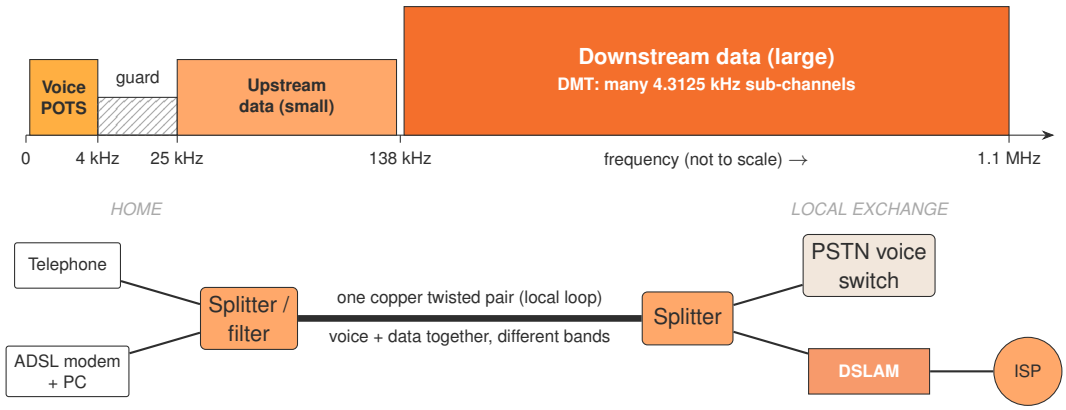


Figure: ADSL frequency split (FDM on one pair) and the splitters that separate voice from data at both ends.

PAST PAPER **ISA2024 Q5b.ii** **[5 marks]**

Describe how ADSL is able to transmit both data and telephone calls over the same twisted pair cable connecting a house to a local exchange, without the two signals interfering with each other.

Model answer

ADSL uses **frequency division multiplexing** on the copper pair:

1. **0–4 kHz**: ordinary voice telephone (POTS), exactly as before.
2. **About 25–138 kHz**: upstream data; **138 kHz–1.1 MHz**: downstream data. A guard band separates voice from data.
3. Because voice and data occupy **different frequency bands**, they do not interfere and can be used at the same time.
4. The data bands are split by **DMT** (discrete multitone) into many small sub-channels, each modulated (QAM) and adapted to the line quality.
5. At the home a **splitter/microfilter** sends low frequencies to the phone and high ones to the ADSL modem; at the exchange another splitter sends voice to the PSTN switch and data to the **DSLAM**, which connects to the ISP.

7.1 ISDN (outline topic)

Draw the frequency bar. It shows both answers at once: the bands explain “no interference”, the bigger downstream block explains “asymmetric”.

ISDN (Integrated Services Digital Network) is an end-to-end **digital, circuit-switched** telephone network carrying voice, data, fax and video over ordinary copper. It uses **B channels** (bearer, 64 kbps, user data) and a **D channel** (signalling/control).

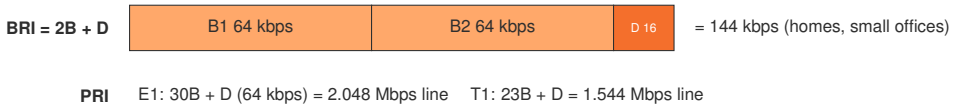


Figure: ISDN channel structure.

ISDN BRI rate

$$\text{BRI} = 2 \times 64 + 16 = 144 \text{ kbps} \quad \leftarrow \textit{192 kbps on the wire with framing}$$
$$\text{E1 PRI} = 30 \times 64 + 64 + 64 = 2048 \text{ kbps} \quad \leftarrow \textit{30 B + D + framing slot}$$

7.2 Terminal handling (outline topic)

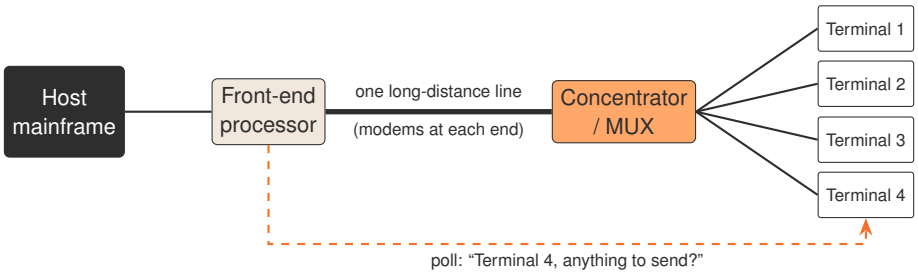


Figure: many dumb terminals share one line to the host through a concentrator; the host polls each in turn.

- **Terminal handling** = connecting many (dumb) terminals to a central host cheaply and controlling who may send.
- **Multiplexer**: fixed sharing (TDM/FDM); output capacity = sum of inputs. **Concentrator**: statistical sharing with buffering; output capacity $<$ sum of inputs because terminals are idle most of the time.
- **Front-end processor**: handles line control, polling and error checking so the host CPU does not have to.
- **Polling**: the host asks each terminal in turn "do you have data?". **Selection**: the host asks a terminal "are you ready to receive?" before sending.

8 Practice questions

PRACTICE Q1 EASY [2 marks]

A noiseless channel has a bandwidth of 3000 Hz and uses 4 signal levels. Find the maximum bit rate.

Answer

Working

$$\text{BitRate} = 2B\log_2 L = 2 \times 3000 \times \log_2 4 \quad \leftarrow \textit{Nyquist: noiseless}$$
$$= 6000 \times 2 = 12\,000 \text{ bps}$$

PRACTICE Q2 EASY [3 marks]

Name the three layers of a fibre optic cable from the centre outwards and give the job of each.

Answer

(1) **Core**: thin glass, higher refractive index, carries the light. (2) **Cladding**: glass of lower refractive index, reflects light back into the core (total internal reflection). (3) **Buffer/jacket**: plastic coating that protects against damage and moisture.

PRACTICE Q3 EASY [3 marks]

State the channel structure and total user rate of ISDN Basic Rate Interface.

Answer

BRI = **2B + D**: two 64 kbps bearer channels for voice/data and one 16 kbps D channel for signalling. $2 \times 64 + 16 = 144$ kbps.

PRACTICE Q4 EASY **[2 marks]**

Differentiate between bit rate and baud rate.

Answer

Bit rate = number of bits sent per second (bps). **Baud rate** = number of signal elements (symbols) per second. Bit rate = baud $\times \log_2 L$, so with 4 levels, 1000 baud carries 2000 bps.

PRACTICE Q5 MEDIUM **[5 marks]**

A telephone line has a bandwidth of 3100 Hz and an SNR of 30 dB. (a) Find its Shannon capacity. (b) How many signal levels are needed to reach this rate?

Answer

Working

$$\text{SNR} = 10^{30/10} = 1000 \qquad \leftarrow \textit{convert dB first}$$

$$C = 3100 \times \log_2(1001) = 3100 \times 9.97 \approx 30\,900 \text{ bps}$$

$$30\,900 = 2 \times 3100 \times \log_2 L \qquad \leftarrow \textit{Nyquist}$$

$$\log_2 L = 4.98 \Rightarrow L \approx 32 \text{ levels} \qquad \leftarrow \textit{round up to a power of 2}$$

PRACTICE Q6 MEDIUM **[4 marks]**

Draw the NRZ-L and Manchester signals for the bit pattern 10110. State one advantage of Manchester.

Answer

NRZ-L: high, low, high, high, low (one level for the whole bit). Manchester (IEEE 802.3): 1 = low-to-high in mid-bit, 0 = high-to-low, so: ↑, ↓, ↑, ↑, ↓, with an extra edge at the bit boundary between two equal bits. **Advantage**: a transition in every bit, so the receiver recovers the clock from the signal (self-synchronising) and there is no DC build-up. (Cost: needs twice the bandwidth.)

PRACTICE Q7 MEDIUM **[5 marks]**

Compare the datagram and virtual-circuit approaches to packet switching.

Answer

Datagram: connectionless, no setup; each packet carries the full destination address and is routed independently, so packets may take different routes, arrive out of order or be lost; robust to node failure; good for short transactions; e.g. IP. **Virtual circuit**: connection-oriented; a route is set up first by a call-request packet; packets carry a short VC identifier and all follow the same route, arriving in order; faster per-packet forwarding, but a node failure breaks the circuit; good for long transfers; e.g. X.25, Frame Relay, ATM.

PRACTICE Q8 MEDIUM **[5 marks]**

Eight sources of 56 kbps are combined by synchronous TDM, one byte per slot, with 1 framing bit per frame. Find the frame size, frame rate and link rate.

Answer

Working

$$\begin{aligned} \text{frame} &= 8 \times 8 + 1 = 65 \text{ bits} && \leftarrow \textit{one byte from every source + 1 framing bit} \\ \text{frame rate} &= 56\,000/8 = 7000 \text{ frames/s} && \leftarrow \textit{each source sends 7000 bytes/s} \\ \text{link rate} &= 65 \times 7000 = 455 \text{ kbps} && \leftarrow \textit{8 × 56 = 448 + 7 kbps framing} \end{aligned}$$

PRACTICE Q9 HARD **[12 marks]**

With the aid of diagrams, discuss three multiplexing techniques used in data communication, giving an application of each and comparing synchronous with statistical TDM.

Answer

Multiplexing lets n signals share one link through a MUX (many-to-one) and DEMUX (one-to-many). (2)
FDM (3): the link bandwidth is divided into frequency bands, one per signal, separated by guard bands; all signals are sent at the same time. Analogue. E.g. radio/TV, cable TV, ADSL. Diagram: stacked bands on a frequency/time graph.
TDM (3): time is divided into slots grouped into frames; each source uses the whole bandwidth for its slot in rotation. Digital. E.g. E1/T1 telephone trunks, GSM. Diagram: A B C | A B C slots.
WDM (2): FDM for light; each signal on its own wavelength in one fibre, combined/split by a prism or grating. E.g. fibre backbones (DWDM, 80+ channels).
Synchronous vs statistical TDM (2): synchronous gives every input a fixed slot in every frame, so idle inputs waste slots; statistical TDM gives slots only to inputs with data, each slot carries an address, so the link can be smaller than the sum of the inputs.

PRACTICE Q10 HARD **[10 marks]**

A home user's telephone line has usable bandwidth from 25 kHz to 1.1 MHz for data and an SNR of 40 dB. (a) Calculate the theoretical maximum data rate. (b) Explain why a dial-up modem on the same line only reaches 56 kbps. (c) Explain why the real ADSL rate is much lower than your answer to (a).

Answer

Working

$$\begin{aligned} B &= 1\,100\,000 - 25\,000 = 1\,075\,000 \text{ Hz} && \leftarrow \textit{bandwidth = high − low} \\ \text{SNR} &= 10^{40/10} = 10\,000 \\ C &= 1.075 \times 10^6 \times \log_2(10\,001) && \leftarrow \log_2 10\,001 \approx 13.29 \\ &\approx 14.3 \text{ Mbps} \end{aligned}$$

(b) A dial-up modem sends through the telephone network's voice channel, which is filtered to about 300–3400 Hz (B ≈ 3.1 kHz), and the exchange samples voice at 8000 × 7–8 bits, giving at most 56 kbps. ADSL bypasses the filter and uses up to 1.1 MHz of the copper pair.
(c) Shannon is a theoretical ceiling; real lines have attenuation that grows with distance and frequency, crosstalk from other pairs, and impulse noise, so DMT turns off or slows the noisy sub-channels. Part of the band is upstream, and coding/framing overhead also takes capacity.

PRACTICE Q11 HARD **[12 marks]**

A company has offices in two cities 300 km apart. Discuss whether it should use a circuit-switched leased line or a packet-switched service for (i) voice calls and (ii) file and e-mail traffic. Support your answer with a diagram and refer to delay, jitter, cost and efficiency.

Answer

Diagram: the circuit vs packet network figure (dedicated path A–S1–S2–B vs numbered packets on different routes). (3)
Voice (4): needs low, *constant* delay and low jitter (under about 150 ms one way). A circuit gives a dedicated path with fixed bandwidth and no queuing, so quality is guaranteed; but bandwidth is wasted in silences and it is paid for all the time. Packet voice (VoIP) is cheaper and acceptable only with QoS (priority queuing, jitter buffers).
Files and e-mail (4): bursty, tolerate delay and jitter, need reliability rather than timing. Packet switching shares links among many users (statistical multiplexing), so it is far cheaper and more efficient; packets reroute around failures; TCP fixes loss and order.
Recommendation (1): a packet-switched service (e.g. MPLS VPN) with QoS for voice, because it carries both types on one link at lower cost; keep a circuit only if guaranteed voice quality is critical.

ONE-PAGE CHEAT SHEET

Media

- Guided = bounded: twisted pair, coax, fibre
- Unguided = unbounded: radio (omni), microwave (line of sight), infrared (no walls)
- UTP $\approx$100 m, twist cancels noise; STP adds a shield
- Coax: conductor, insulator, braid shield, jacket
- Enterprise: UTP to desks, fibre backbone, Wi-Fi APs
- Distance: UTP 100 m, coax 500 m, fibre 2–100+ km

Fibre

- LED/laser $\rightarrow$ core $\rightarrow$ photodetector
- Core index $>$ cladding index; angle $>$ critical $\Rightarrow$ total internal reflection
- SM: thin core, laser, long; MM: thick, LED, short
- Beats copper: bandwidth, low attenuation, EMI-immune, secure, light

Signals

- Analogue continuous; digital discrete, regenerable
- Impairments: attenuation, distortion, noise (thermal, induced, crosstalk, impulse)
- Sync = shared clock, blocks; async = start/stop bits
- Nyquist $2B \log_2 L$; Shannon $B \log_2(1 + \text{SNR})$
- SNR = $10^{\text{dB}/10}$; bit rate = baud $\times \log_2 L$

Modulation vs encoding

- Modulation: carrier, analogue out: ASK, FSK, PSK, QAM
- Encoding: voltage levels: NRZ, Manchester, AMI
- Manchester: 0 = high$\rightarrow$low, 1 = low$\rightarrow$high

Multiplexing

- MUX many-to-one; DEMUX one-to-many; link + channels
- FDM: bands + guard bands, analogue, radio/ADSL
- TDM: time slots + framing, digital, E1/GSM
- Stat TDM: slots on demand; WDM: wavelengths on fibre
- GSM = FDMA+TDMA, SIM, hard handoff; CDMA = codes, soft handoff

Switching

- Circuit: dedicated path, setup/data/teardown, PSTN, ISDN; fixed delay
- Message: whole message store-and-forward, telegraph
- Packet: packets share links, IP, X.25, Frame Relay; jitter
- Packet = highest latency & jitter (queues, different routes)

Datagram vs VC

- Datagram: no setup, full address, own route, may reorder (IP)
- VC: call setup, short VCI, same route, in order (X.25, ATM)

ADSL

- Asymmetric: downstream $\gg$ upstream (8 vs 1 Mbps)
- FDM: voice 0–4 kHz, up 25–138 kHz, down 138 kHz–1.1 MHz
- DMT sub-channels; splitter at home, splitter + DSLAM at exchange

ISDN: digital, circuit-switched. BRI 2B+D = $2 \times 64 + 16 = 144$ kbps; PRI 30B+D (E1, 2.048 Mbps), 23B+D (T1).

Terminal handling: MUX (fixed) vs concentrator (statistical, buffers); front-end processor; polling (“send?”) and selection (“ready to receive?”).